

The Becker–Bergweiler question

What it settles, what it does not, and how we might close the gap

Briefing note for Chapter 2

6 August 2026

Abstract

Chapter 2 argues that the quasi-stationary constant $A(p)$ of a subcritical binary Galton–Watson process has no elementary closed form. The argument runs through a theorem of Becker and Bergweiler on the hypertranscendence of Koenigs linearisations. That theorem is correct, its hypotheses are met, and the algebra connecting it to $A(p)$ has been verified exactly. But the theorem is a statement about a *function of the dynamical variable at fixed parameter*, whereas the chapter wants a statement about a *single value*, and about the *map* $p \mapsto A(p)$. Those are three different assertions, and only the first is proved. This note sets out the chain, isolates the gap precisely, records an unresolved bibliographic problem (three candidate papers, one of which may not exist), and assesses four routes to a resolution — including the observation that the most obvious route buys less than it appears to.

1 The object

For the binary branching process with survival recursion $S_{n+1} = 2pS_n - pS_n^2$, $S_0 = 1$, write

$$A(p) := \lim_{n \rightarrow \infty} \frac{S_n}{(2p)^n} \in (0, \tfrac{1}{2}], \quad p \in (0, \tfrac{1}{2}). \quad (1)$$

Its reciprocal $1/A(p)$ is the limiting conditional mean $\lim_n \mathbb{E}[Z_n \mid Z_n > 0]$, which is why it matters. The substitution $S_n = 2w_n$ turns the recursion into the logistic map

$$w_{n+1} = f_r(w_n) = r w_n(1 - w_n), \quad r = 2p, \quad w_0 = \tfrac{1}{2}, \quad (2)$$

and $r = 2p$ is exactly the mean offspring number, so the subcritical window is $r \in (0, 1)$. Note that $w_0 = \frac{1}{2}$ is precisely the critical point of f_r .

2 The chain of reasoning

The chapter’s argument has four links. Links 1–3 are established; link 4 is where the question lives.

Link 1: Koenigs. For $0 < |\lambda| < 1$ with $\lambda = f'(0) = r$, Koenigs’ theorem supplies a unique analytic ψ_r with $\psi_r(f_r(z)) = r\psi_r(z)$ and $\psi_r'(0) = 1$, and ψ_r extends uniquely to the whole basin of attraction of 0. This gives

$$A(p) = 2\psi_r\left(\tfrac{1}{2}\right), \quad r = 2p. \quad (3)$$

Status: verified to ≥ 117 significant digits by four mechanistically different computations, one of which (the exact-rational Schröder series) is logically independent of the limit definition. The extension to the basin is load-bearing and not optional: the Taylor germ at 0 has radius exactly $(1 - r)/r$, so $w_0 = \frac{1}{2}$ lies outside it for all $p \geq \frac{1}{3}$ — most of the interesting range.

Link 2: the conjugacy. With $h(w) = -rw + r/2$,

$$h \circ f_r \circ h^{-1}(z) = z^2 + c, \quad c = \frac{2r - r^2}{4}, \quad (4)$$

an exact identity in $\mathbb{Q}(r)[z]$. The multiplier is preserved, and c maps $(0, 1)$ bijectively onto $(0, \frac{1}{4})$ — the real segment of the Mandelbrot set from the root of the main cardioid to its cusp. Since $h(\frac{1}{2}) = 0$,

$$A(p) = -\frac{2}{r} \psi_{P_c}(0) : \quad (5)$$

the quasi-stationary constant is the Koenigs coordinate of the critical point of a quadratic polynomial. The transfer $\psi_r(w) = -\psi_{P_c}(h(w))/r$ makes ψ_r differentially algebraic exactly when ψ_{P_c} is — the step that licenses applying a theorem about $z^2 + c$ to f_r , and which neither source document stated.

Link 3: Becker–Bergweiler.

Theorem 1 (Becker–Bergweiler, 1995). *The Koenigs linearisation of a rational map at a fixed point with $0 < |\lambda| < 1$ satisfies an algebraic differential equation if and only if the map is conformally conjugate to a monomial or to $\pm$ a Chebyshev map.*

Every quadratic polynomial is affinely conjugate to a unique $z^2 + c$; the monomial case is $c = 0$ and the Chebyshev case is $c = -2$. By (4) the branching window $c \in (0, \frac{1}{4})$ meets neither. The exclusion is doubly safe: z^2 has multipliers $\{0, 2\}$ and $z^2 - 2$ has $\{4, -2\}$, so neither exceptional map even possesses a fixed point with $0 < |\lambda| < 1$. Hence, for every fixed $r \in (0, 1)$, the function ψ_r satisfies no algebraic differential equation. Since every elementary function in the Liouville sense is differentially algebraic, ψ_r is not elementary.

Link 4: the conclusion — and the gap. The chapter would like to conclude that $A(p)$ has no closed form. It cannot, and saying why is the point of this note.

3 The gap, stated precisely

Theorem 1 concerns the function $z \mapsto \psi_r(z)$ at fixed r . Three distinct assertions are in play, and only the first has been proved:

- (A) For each fixed $r \in (0, 1)$, the function $z \mapsto \psi_r(z)$ is not elementary. **PROVED**
- (B) The single value $\psi_r(\frac{1}{2})$, i.e. $A(p)$ at one p , has no closed form. **NOT PROVED**
- (C) The map $p \mapsto A(p)$ on $(0, \frac{1}{2})$ is not an elementary function of p . **NOT PROVED**

(A) does not imply (B) or (C), and the counterexample is elementary in both senses of the word: the gamma function is hypertranscendental by Hölder’s theorem, and yet $\Gamma(\frac{1}{2}) = \sqrt{\pi}$. A hypertranscendental function may take perfectly elementary values at perfectly ordinary points. Nor is there any general principle carrying non-elementarity in the dynamical variable z across to non-elementarity in the parameter r : these are different variables of a two-variable object $\psi_r(z)$, and differential algebraicity in one says nothing about the other.

An earlier draft asserted (B) and (C) outright — “for the whole parameter range $p \in [0, 0.5)$ there is no closed form representation of $\psi_p(1/2)$ in terms of p , and hence neither for $A(p)$ ” — and that sentence has been withdrawn to a footnote. The chapter’s present position is the hedged one: no elementary closed form is known, and none is expected, with the structural reason given; whether one exists is stated as open.

Question 1. Is $A(p)$ expressible in closed form — either as a value at fixed p , or as a function of p on $(0, \frac{1}{2})$?

4 An unresolved bibliographic problem

Compounding the mathematical gap is a citation problem that must be settled before the chapter goes out. Three papers are in circulation, and they are not the same paper.

#	Title (abbreviated)	Venue	Status
1	Hypertranscendency of conjugacies in complex dynamics	Math. Ann. 301 (1995) 463–468	confirmed
2	Transcendency of local conjugacies ... and of their values	Manuscripta Math. 81 (1993) 329–338	confirmed
3	Hypertranscendency of local conjugacies and semiconjugacies	Proc. AMS 123 (7) (1995) 2199–2205	unconfirmed

Paper 1 is confirmed by DOI 10.1007/BF01446639, appears in Bergweiler’s own publication list and in several independent bibliographies. It is the citation used in the author’s own `references.bib`, and it is what the chapter currently cites.

Paper 2 is confirmed via EUDML record 155904, authors Paul-Georg Becker and Walter Bergweiler, with keywords *Mahler’s method*, *transcendence*, *Böttcher functions*. It concerns transcendency of the *values* of local conjugacies at algebraic points.

Paper 3 is the citation used by the two machine-generated source documents, and it could not be confirmed to exist. Its title looks like a splice of fragments from real, adjacent titles: “hypertranscendency of ... conjugacies” from paper 1, “local conjugacies” from paper 2, and “semiconjugacies of rational functions” from a genuine but *different author’s* paper (Eremenko) that surfaces in the same searches. This is stated as a pattern, not a proof — the AMS archive returned 403 on direct fetches, so absence of confirmation is not proof of non-existence. Nothing in the chapter cites it, and nothing should until someone confirms it exists.

5 Four routes to a resolution

Route 1 — Read paper 2, and check whether its argument transfers

Paper 2 is about transcendency of the *values* of local conjugacies at algebraic points, with the same monomial/Chebyshev exceptional structure. Since $\frac{1}{2}$ is algebraic, this is the right neighbourhood for gap (B).

Caution, in two parts. First, paper 2 concerns Böttcher conjugacies near infinity for polynomials of equal degree, not Koenigs linearisation at a finite attracting fixed point. It is adjacent, not identical; someone must read it and judge whether the argument transfers. Second — and this is the part that is easy to miss — *even complete success here buys less than it appears to*. A theorem that $\psi_r(\frac{1}{2})$ is transcendental for algebraic r would not establish that $A(p)$ has no closed form: $\Gamma(\frac{1}{2}) = \sqrt{\pi}$ is transcendental and has a closed form. Transcendence and closed-formness are different properties. Route 1 would let the chapter say something sharper and true about the arithmetic nature of $A(p)$, which is worth having; it would not answer Question 1.

Cost: one afternoon in a library. *Payoff:* a genuine strengthening, but of a different claim than the one in question.

Route 2 — Attack (C) directly: differential algebraicity in the parameter

This is the statement the chapter actually wants. The object is $A : (0, \frac{1}{2}) \rightarrow (0, \frac{1}{2}]$, real-analytic, and the question is whether it satisfies an algebraic ODE *in* p . Becker–Bergweiler is silent here because it is a theorem about the wrong variable.

This is the mathematically honest target and also the hardest. It is a question about how the Koenigs linearisation varies with its parameter, evaluated along the critical-point section — and the critical-point section is exactly what makes the quadratic family’s parameter dependence delicate, which is the same delicacy that makes the Mandelbrot set what it is. There may be literature on the differential algebraicity of parametrised linearisers; it should be searched before anyone attempts a proof.

Cost: a literature search, then possibly a research project. *Payoff*: would settle Question 1 in the direction the chapter expects.

Route 3 — Try to *find* a closed form, adversarially

If (B) or (C) is false, the fastest way to learn that is to look for the formula. The brief at `build/PROMPT_attack_A_of_p.md` sets this up: high-precision values of $A(p)$ at many p , inverse symbolic search and integer-relation detection (PSLQ) at 200+ digits, numerical tests of differential algebraicity in p , and a systematic sweep of special-function representations. A hit would overturn the chapter’s position; a thorough miss is not a proof but is real evidence, and would let the chapter say the search was conducted rather than merely asserting that none is known.

Prerequisite: `mpmath` is not installed in the working environment and is required for PSLQ. *Cost*: compute time. *Payoff*: asymmetric — cheap, and either outcome is publishable content.

Route 4 — Keep the hedge, and say why it is a hedge

The chapter’s current position. Cite paper 1 for (A), state (B) and (C) as open, and give the structural reason for the expectation: $A(p)$ is the Koenigs coordinate of the critical point of a quadratic polynomial whose parameter runs over $(0, \frac{1}{4})$, a family for which the linearisation is provably non-elementary at every parameter.

This is defensible exactly as written and costs nothing. It is also the correct fallback if Routes 1–3 do not land before submission. Its weakness is that a reader may ask why the chapter stops where it does, so the footnote explaining the values-versus-functions distinction is doing necessary work and should not be cut.

6 What must be decided

1. **Which paper does the chapter cite?** Paper 1 is confirmed and supports exactly claim (A), which is what the chapter uses. Recommendation: cite paper 1 alone unless Route 1 is carried out, in which case cite both. Do not cite paper 3.
2. **Is Route 1 worth an afternoon?** It sharpens a claim the chapter does not currently make, rather than closing the gap it currently flags. Worth doing if the arithmetic nature of $A(p)$ is of interest in its own right; not worth doing if the only goal is Question 1.
3. **Is Route 3 worth the compute?** Recommendation: yes. It is the cheapest of the three active routes and the only one with a chance of a decisive negative result.
4. **The remaining citation gap.** “Every elementary function in the Liouville sense is differentially algebraic” is stated in the chapter with a `\todo`. It is standard and true, but it is a theorem and the machine-generated sources gave it as an unreferenced remark. A survey on transcendently transcendental functions is the likely home for a citable statement; this has not been verified and someone should check it rather than taking this note’s word for it.

Summary

The Becker–Bergweiler theorem is real, correctly stated, correctly applied, and its hypotheses are exactly met — verified by exact computation, not assumed. What it proves is that the linearising function is not elementary at any parameter in the branching window. What it does not prove, and what the chapter formerly claimed it proved, is that the number $A(p)$ or the map $p \mapsto A(p)$ has no closed form. The distinction is $\Gamma(\frac{1}{2}) = \sqrt{\pi}$. The chapter now states this as open, which is the honest position; the routes above are the ways to stop it being open, and the cheapest of them (Route 3) has the useful property that both outcomes are worth having.